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GATE ECE · NETWORK THEORY · CHAPTER-WISE PYQS

State Equations For Networks

GATE ECE Network Theory · Previous Year Questions with Answer Key

4	4	1996 – 2020	English
QUESTIONS	PAPERS COVERED	YEARS	LANGUAGE

INSIDE THIS PDF

Marks 2	1 question
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Answer key	at the end



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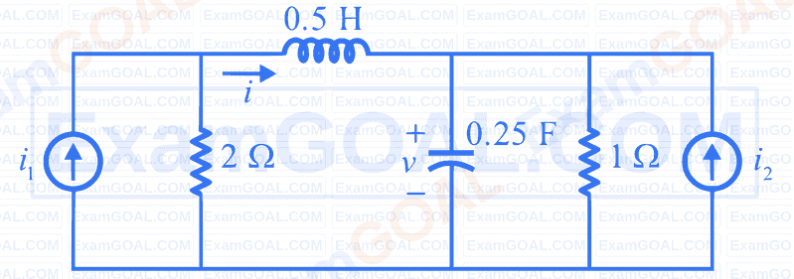
1. **Solve**
   Print it, or work through it on screen.
2. **Check**
   The answer key is on the last pages.
3. **Understand**
   Scan the QR code beside any question for its full solution.

Marks 2 1 questions

Q1. GATE ECE 2020

Open solution

For the given circuit, which one of the following is the correct state equation?



(A) $\frac{d}{dt}\begin{bmatrix} v \\ i \end{bmatrix} = \begin{bmatrix} -4 & -4 \\ -2 & -4 \end{bmatrix}\begin{bmatrix} v \\ i \end{bmatrix} + \begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix}\begin{bmatrix} i_1 \\ i_2 \end{bmatrix}$

(B) $\frac{d}{dt}\begin{bmatrix} v \\ i \end{bmatrix} = \begin{bmatrix} -4 & 4 \\ -2 & -4 \end{bmatrix}\begin{bmatrix} v \\ i \end{bmatrix} + \begin{bmatrix} 0 & 4 \\ 4 & 0 \end{bmatrix}\begin{bmatrix} i_1 \\ i_2 \end{bmatrix}$

(C) $\frac{d}{dt}\begin{bmatrix} v \\ i \end{bmatrix} = \begin{bmatrix} 4 & -4 \\ -2 & -4 \end{bmatrix}\begin{bmatrix} v \\ i \end{bmatrix} + \begin{bmatrix} 0 & 4 \\ 4 & 4 \end{bmatrix}\begin{bmatrix} i_1 \\ i_2 \end{bmatrix}$

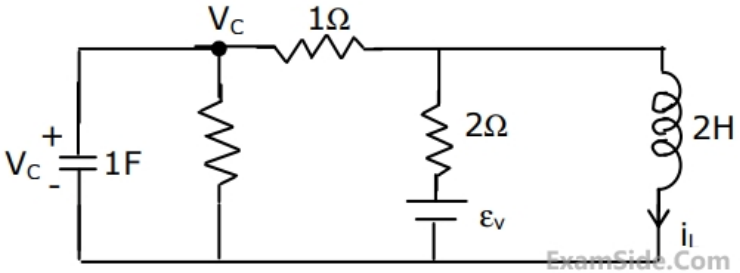
(D) $\frac{d}{dt}\begin{bmatrix} v \\ i \end{bmatrix} = \begin{bmatrix} -4 & -4 \\ -2 & 4 \end{bmatrix}\begin{bmatrix} v \\ i \end{bmatrix} + \begin{bmatrix} 4 & 4 \\ 4 & 0 \end{bmatrix}\begin{bmatrix} i_1 \\ i_2 \end{bmatrix}$

Marks 5 3 questions

Q2. GATE ECE 2000

Open solution

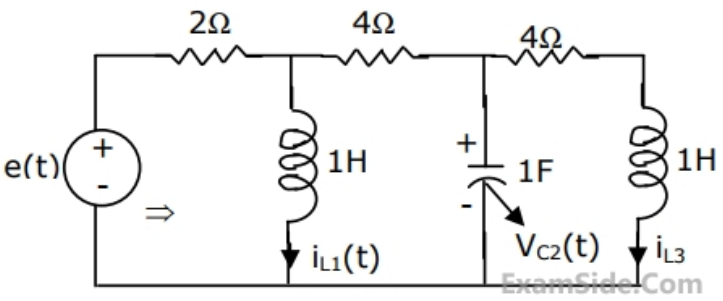
For the circuit in Fig., write the state equations using v_C and i_L as state variables.



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Q3. GATE ECE 1997

Open solution



For the circuit shown in Fig. choose state variables $X_1,\ X_2,\ X_3$ to be $i_{L1}(t),\ v_{C2}(t),\ i_{L3}(t)$
(a) Write the state equations

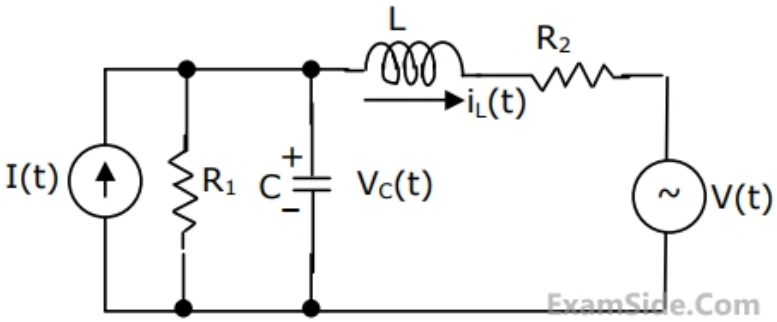
$$\begin{bmatrix}\dot{X}_1\\ \dot{X}_2\\ \dot{X}_3\end{bmatrix} = A\begin{bmatrix}X_1\\ X_2\\ X_3\end{bmatrix} + B[e(t)]$$

(b) If e(t) = 0, t $\geq$ 0, $i_{L1}(0) = 0,\ v_{C2}(0) = 0,\ i_{L3}(0) = 1A$, then what would the total energy dissipated in the registors in the interval $(0, \infty)$ be

Q4. GATE ECE 1996

Open solution

Refer to the circuit shown in Fig.



Choosing the voltage $v_C(t)$ across capacitor, and the current $i_L(t)$ through the inductor as state variable,i.e.,

$$[x(t)] = \begin{bmatrix}v_C(t)\\ i_L(t)\end{bmatrix}$$

Write the state equation in the form $\frac{d[x(t)]}{dt} = [A][x(t)] + [B][u(t)]$ and find [A], [B] and [u(t)].

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Answer Key

Marks 2

1. **B**		

Marks 5

2. **Solve it.**	3. **(a) Solve it.** **(b) 0.5 J**	4. **Solve it.**

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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